



INSTALLATION, OPERATIONS, AND MAINTENANCE MANUAL FOR:

LACO Custom Vacuum System Revision A: February 2026



Project Information

Customer:	University of Illinois
Model:	FCT3048ELSSSE-1P35531
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1. Introduction

1.1. Contact Information

For technical assistance or to order replacement parts call, fax or email from the information listed below.

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1.2. About the Manual

The purpose of this manual is to provide instructions for the installation, operation, troubleshooting, maintenance, and documentation of the **LACO Custom Vacuum System**. The sections of the manual may refer to other binders or manuals not enclosed under this cover as well as software that will be included in the appendix. The entire manual contents will also be saved on CD or USB drive and included in the appendix for electronic storage.

1.3. Safety & Warning Messages

See below for the range of warning messages used throughout this manual. The lower text in these messages indicates the severity range of the message group.



DANGER

Imminent threat of danger resulting in death or severe injuries.



WARNING

Dangerous situation resulting in major injuries.



CAUTION

Dangerous situation resulting in minor injuries.

NOTICE

Dangerous situation resulting in damage to the equipment, property, or environment.

2. Safety

This section describes information that is important for safe equipment operation.

2.1. Intended Use

The LACO Vacuum System is designed to create a vacuum environment inside the chamber while providing heating and cooling to the intended surface.

- You must install, operate and service this device only in compliance with these operating instructions.
- Adhere to the storage and usage recommendations.

2.2. Owner Requirements

2.2.1. Safe Operation

- Operate the device only when it is in technically perfect working order.
- Operate the device only as specified in a safety-conscious and hazard-conscious manner and in compliance with these operating instructions.
- Comply with the following regulations and monitor their compliance.
 - Intended use
 - Safety and accident prevention regulations
 - International, national, and local standards and guidelines
 - Additional provisions and regulations that are specific to the unit
- Use only original parts or parts approved by the manufacturer.
- Keep these operating instructions available at the equipment location.

2.2.2. Personnel Qualifications

- Allow only qualified service technicians to work with and on the device. The qualified service technicians must have received training on the device.
- Allow personnel in training to work with and on the device only under the supervision of trained qualified service technicians.
- Make sure that the authorized personnel have read and understood the operating instructions and all other applicable documents, especially the information regarding safety, maintenance, and repairs, before starting work.
- Never perform maintenance on the system with the power on.

2.3. Operator Requirements

- Read, observe, and follow the information in these operating instructions and the working instructions created by the owner, especially the safety instructions and warnings.
- Carry out any work only based on the complete operating instructions.
- If you have any questions regarding operation or maintenance that you cannot find answers to in these instructions, then please contact LACO customer service.
- When using this equipment, be sure to follow the safety procedures outlined by your facility. These safety procedures should cover the two primary types of hazard training: (1) equipment hazards and (2) facility-related hazards.
- Never override or bypass any hardware or software interlocks.
- All safeguard devices must be in place when equipment is in operation. Operators, set-up operators, helpers or installation personnel should not alter, remove, or disable safety equipment.
- The Vacuum System must be operated according the specification in this manual, otherwise the protective measures may be impaired

2.4. Dangers

2.4.1. Electrical Power



Considerable voltages arise inside the device. Touching parts where electrical voltage is applied can result in death. Only trained and qualified personnel should work inside the electrical panel.

- Disconnect the device from the electrical power supply prior to any installation and maintenance work. Make sure that the electric power supply is reconnected with authorization. Testing live parts results in danger to life.
- The device contains electric components that can be damaged from high electric voltage.
- Before connecting the device to the power supply, make sure that the supply voltage specified on the device is the same as the local power supply.
- Keep all liquids away from electrical panels, and connections.

2.4.2. Air Pressure

This system operates using pressurized air. Visually inspect all tubing and tubing connection for damage before use. Replace any damaged tubing or tubing connection before operating



WARNING

Never exceed the recommended pressures outlined in this document and identified on the system.

2.4.1. Process Gas Pressure

This system may use low-pressure nitrogen to backfill after testing. Visually inspect all tubing and tubing connection for damage before use. Replace any damaged tubing or tubing connection before operating.



WARNING

Never exceed the recommended pressures outlined in this document and identified on the system.

2.4.2. Vacuum Forces

NOTICE

There is strong suction on the inlet port of the vacuum chamber. If parts of the body or other objects are near the inlet; they may be sucked in. Always keep parts of the body and other objects away from the vacuum inlet.

2.4.3. Hot Surfaces

- Do not operate the heater unless the system is under vacuum.
 - There is an advanced user setting which will prevent the heaters from turning on until a certain vacuum level has been reached. See HVC-3500 manual for more information.
- Do not vent the system until the heater has reached ambient conditions
 - There is an advanced user setting which will prevent the vent valve from opening when the system is above a given temperature. See HVC-3500 manual for more information.

**WARNING**

Hot Surfaces. Do not touch heated shelves or inner chamber walls unless temperature is at ambient conditions.

**CAUTION**

Hot Surfaces. Chamber outside may become hot during regular operation.

2.4.1. Pressurized Chamber

This chamber is only capable of sustaining vacuum and is not capable of acting as a pressure vessel.

**DANGER**

Do not pressurize chamber. Pressurizing chamber could lead to door clamp failure and/or chamber rupture

2.4.2. Moving Parts

- Keeps hands or other body parts clear of any moving parts.
- Lock out power and gases when performing service or maintenance

2.4.3. Dangers from Liquids and Vapors

- Liquids and chemical substances can damage the device.
- Adhere to the restrictions of use.
- Do not suck up liquids with the device.
- Never contact toxic, caustic, microbiological, explosive, radioactive or other harmful substances with the device.
- Only clean the device using mild household detergents or alcohol. Hydrogen and air form a highly explosive mixture.
- Do not use tracer gases with a hydrogen concentration > 5%.
- Only use the device outside potentially explosive areas.
- No smoking. Do not subject the device to open fire and avoid sparking.
- If condensable gases or vapors are measured, there is a risk of pump corrosion and damage.

2.5. Equipment Safety

The rest of this section describes equipment safety features:

- Process Faults/Interlocks
- Lockout/Tag Out Information
- Equipment Safety Labels

2.5.1. Pressure Relief Valve

The vacuum chamber includes a pressure relief which prevents the chamber from becoming pressurized while filling with the process gas.



DANGER

Never adjust or remove pressure relief valves.

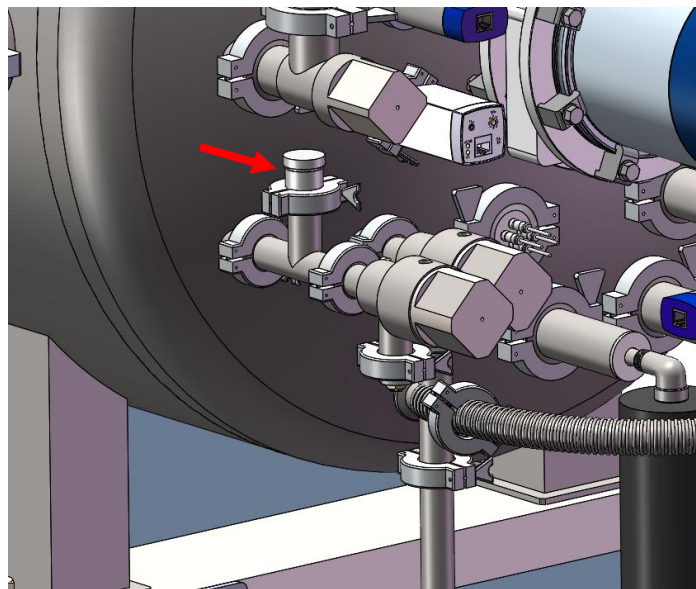


Figure 2-1. Chamber Pressure Relief Valve

2.5.2. Process Faults/Interlocks

The process fault interlocks are latched into the logic controller (PC) memory. When a fault occurs, the user interface will display the alarm condition and the light tower will flash red. Pressing the RESET button will clear the fault once the fault condition has been fixed. If the indicator on the screen remains illuminated, then one or more faults are still active. Refer to [Section 7](#) Trouble Shooting.

CAUTION

The interlock circuit does not remove hazardous voltage from the system. Only authorized, qualified, and trained personnel should service this equipment.

2.5.3. Lockout / Tagout (LOTO)

Preliminary

Before installation or servicing the system, the facility's power source to the system must be de-energized to prevent serious injury to personnel and equipment. Additionally, all pressurized gas sources (compressed air, helium, nitrogen, etc.) must be de-energized and removed from the system before installation or servicing can occur. An authorized employee representing the facility installing the system must follow approved company guidelines and lockout or use suitable means to prevent re-energizing the electrical system and/or pressurized gas sources during installation or servicing.

Definitions

- **Lockout:** the placement of a lockout device on an energy isolating device, in accordance with established company procedures, ensures that the energy isolating device and the equipment being controlled cannot be operated until the lockout device is removed.
- **Tag Out:** a prominent warning device such as a tag and a means of attachment, which can be securely fastened to an energy isolating device in accordance with established company procedure, ensures that the energy isolated device and the equipment being controlled may not be re-energized or operated until the tagout device is removed.

This table lists the lockout/tagout information for the system.

Table 2-1. Lockout/Tagout	
Energy Type: Electrical	
Hazard:	Electrocution, electrical burns, and shock
Magnitude:	See main system label
Control Method:	Main Power
Shutdown Procedure	
Switch off disconnect switch on electrical enclosure at back of system	

Energy Type: Pressurized Compressed Air	
Hazard:	Flowing high pressure gas, explosion
Magnitude:	See main system label
Control Method:	3-way shut-off valve
Shutdown Procedure	
Switch off 3-way shut-off valve on main control air filter regulator module on connection panel or remove compressed air line from system	

Energy Type: Pressurized Gas (Helium, Air, Nitrogen)	
Hazard:	Flowing high pressure gas, explosion
Magnitude:	See main system label
Control Method:	Turn off gas source bottles and vent gas lines.
Shutdown Procedure	
Turn off gas source bottles and vent gas lines.	

Machine Shutdown

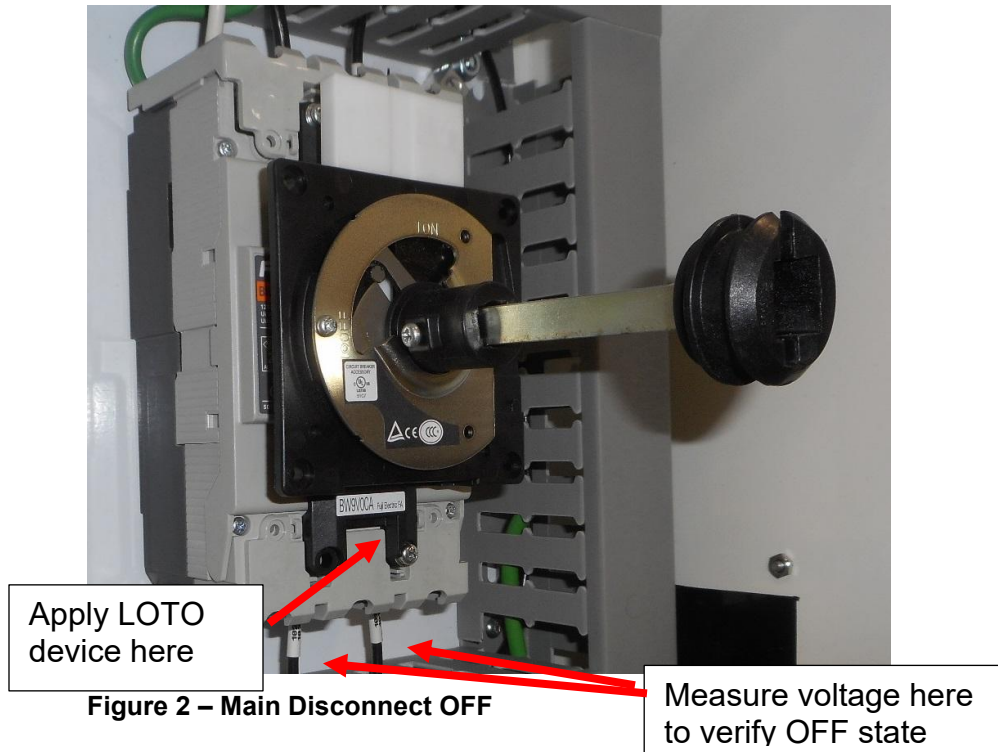
Perform the following sequence of events in the order listed for electrical energy isolation of the tool:

1. Notify personnel in the area that you are going to shut down the equipment.
2. Locate the main incoming power disconnect switch-actuating handle.
3. Move the actuating handle to the "OFF" (down) position.
4. Verify the voltage is off by opening the door and measuring the voltage between the terminals on the bottom of the main disconnect. (See Figure 2 – Main Disconnect)
5. Apply the locking energy isolation device (lock) through the hole in the actuating handle and secure the lock.
6. Verify that the tool has been isolated and de-energized by attempting to turn the main power disconnect back to the "ON" position and/or by pressing the machine start button. The machine power must not be reapplied and/or the machine must not start.

Machine Start-Up

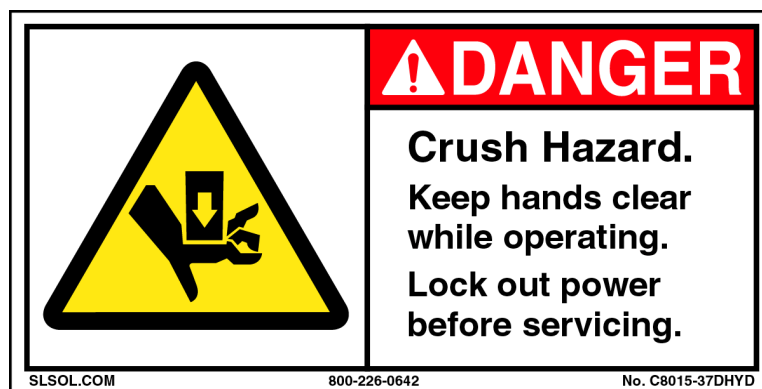
Perform the following sequence of events in the order listed for electrical re-energization of the tool:

1. Ensure that all hand tools are removed from the equipment and that it is ready for start-up.
2. Notify personnel in the area that you are going to start-up the equipment.
3. Open the lock and remove the locking energy isolation device (lock) from the hole in the actuating handle of the Main Disconnect Switch.
4. Close the enclosure door.
5. Move the actuating handle to the "ON" (up) position.
6. The machine should power on and be ready to start after HMI boot process.



2.5.4. Safety Labeling

Crush Hazard. Denotes a moving part of the machine that is interlocked but should be serviced only with the power disconnected with LOTO equipment in place.





Hazardous Voltages. Identifies enclosures with voltages over 24VDC that should be serviced only with the power disconnected with LOTO equipment in place.



3. Scope of Delivery, Transportation & Storage

3.1. Shipment items

The following is included in the shipment:

- LACO Custom TVAC (1ea)
- Spare chamber seal (1ea)
- Spare pirani/capacitance gauge (1ea)
- Spare cold cathode gauge (1ea)
- Spare desiccant vent (1ea)

3.2. Transport and Location

NOTICE

Transport the system upright and level.

The LACO Vacuum System will be shipped on a pallet and in this state, should only be transported with a forklift or heavy-duty pallet jack. Remove the system from the crate and inspect system for any signs of damage (dented panels, paint scratches, etc.). Any damage to the system should be reported to the carrier immediately. Be careful to not damage the casters located under the leak test system when using forklift or pallet jack.

Once the system has been removed off the pallet the unit can be moved by the methods listed below.

- Rolled on the casters. If using this method, ensure the leveling feet are fully retracted. The system is very heavy and will need to be pushed by multiple individuals.
- Forklift or pallet jack. Ensure the forks are securely under the system frame before lifting and moving.

! CAUTION

Heavy Object. When lifting or moving the system, follow safe heavy object handling methods to prevent injury.

Take care when transporting the system. The system contains sensitive instrumentation and vibrations and shocks should be minimized to avoid equipment damage.

Locate the unit with easy access to required utilities (reference section 4.1). Once the unit is located, twist the caster leveling feet until they contact the

ground. The main electrical cabinet is on the right side of the system, so this area must have maintenance access.

3.3.Storage and usage conditions

The system should always be stored and used in a dry, clean environment, free of excess dust and debris. Care should be taken to avoid getting dust and debris inside the controller and vacuum chamber. Always store with the chamber closed to help avoid contamination.

If a turbo pump has been stored for more than 12 months, a ball-bearing run-in procedure for smooth ball bearing running must be carried out when putting into operation again. See the pump user manual for instructions on how to perform this procedure.

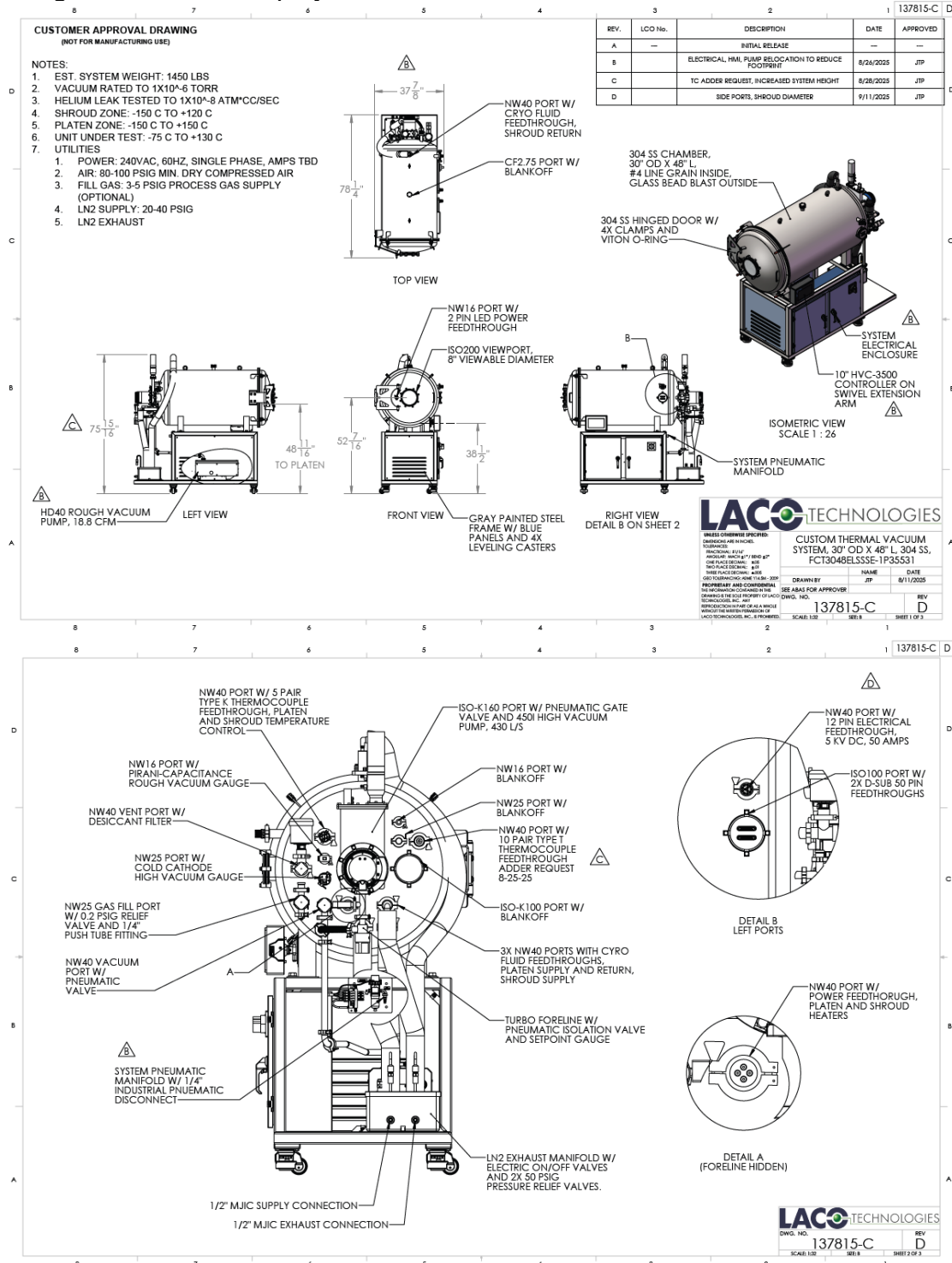
3.4.Decommissioning

At the end of the product life, dispose of all electronic equipment per local laws/regulations.

Some parts of the system are made of recyclable materials. This option may be used to avoid waste and protect the environment.

4. Equipment Description

This system is designed to create a 2 zone thermal vacuum environment for testing satellites with deployable features.



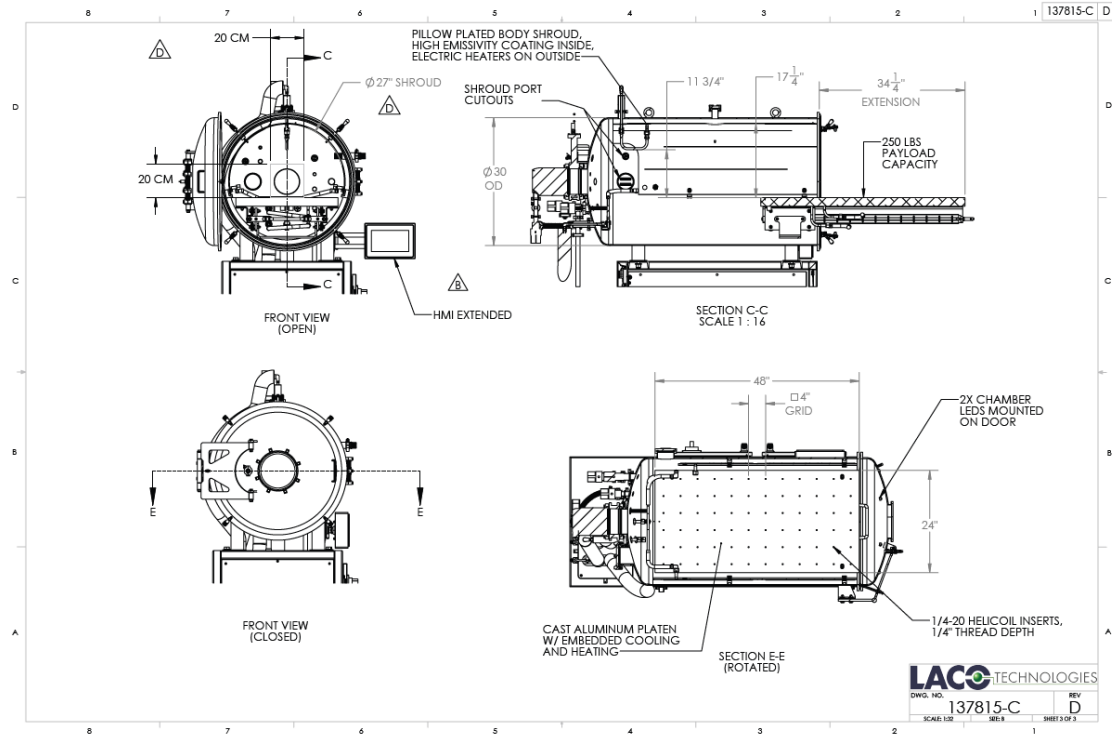


Figure 4-1. Vacuum System

4.1. System Specifications

Table 4-1. System Specifications

System Weight	1450 lbs.
System Dimensions	38" [W] X 79" [D] X 76" [H]
Electrical Requirements	208-240 VAC 40 Amps 60 Hz Single Phase
Air Supply Requirements	80-120 psi
Process Gas Supply	< 5 psi
Liquid Nitrogen Supply	20-40 psi
Maximum Rated Pressure	0.5 psig MAX
Platen Temperature Range	-150 °C to 150 °C
Shroud Temperature Range	-150 °C to 120 °C

CAUTION

Do not exceed pressure ratings and ensure all customer supplied gas related components are rated for the maximum pressure.

5. Installation Procedures

5.1. Unpacking

Remove the system from packaging and inspect for damage. Any damage to the system should be reported to the carrier immediately.

5.2. Location

Locate the system as required. Lower the leveling pads until the unit rests evenly on the floor.

5.3. Electrical

Connect electrical power to the main electrical disconnect. See 4.1 System Specifications for electrical power requirements. See HVC-3500 manual for more information.

5.3.1. Main System Power

The Main electrical power line must be connected using the specifications outlined in cord and plug should not be used as a main disconnecting means. The system can disconnect power and prove Lock-Out / Tag-Out (LOTO) functionality using the main disconnect switch.

- Lockout and Tagout facility power connection switch or equivalent before installing system.
- Open the back door by rotating the electrical disconnect handle to the OFF position. This will allow access to the electrical connections.
- Route the wires from the electrical source into and through the available cord grip located on the side of the main enclosure.
- Close the door and secure.
- Check ground continuity on cabinet to any facility ground.
- Remove facility power Lockout/Tagout.
- Restore facility power supply to vacuum system.

CAUTION

Depending on installation and routing of supply lines, a trip hazard may be created. It is the responsibility of the facility installation personnel to eliminate or minimize all trip hazards or identify such hazards with visual warning signs.

5.4. Air Supply

Connect air pressure to the system 1/4" stem quick connect supply on the rear of the system. See for air supply requirements. See HVC-3500 manual for more information.

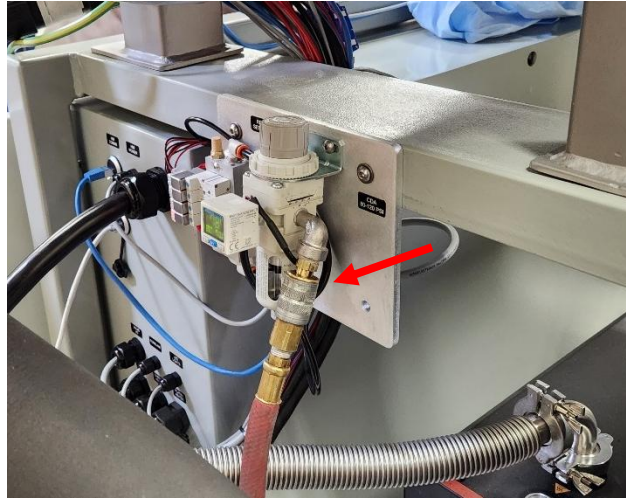


Figure 5-1. Vacuum System

5.5. Process Gas

Connect process gas to the 1/4" push tube fitting on the chamber fill valve. See section 4.1 for air supply requirements. See HVC-3500 manual for more information.

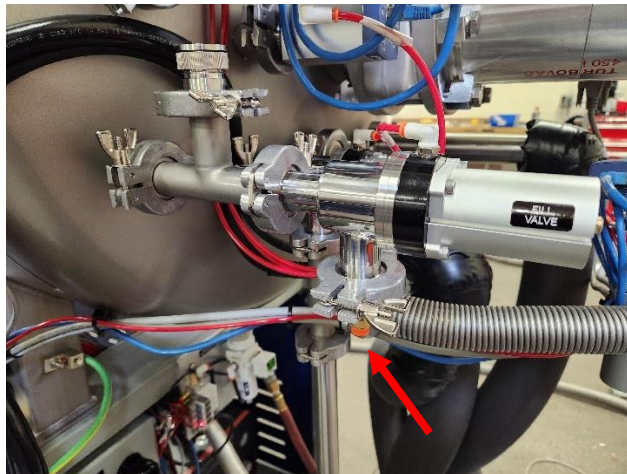


Figure 5-2. Vacuum System

5.6. Liquid Nitrogen

Connect LN2 supply to system LN2 manifold supply via ½" JIC fitting. Connect LN2 exhaust line to manifold outlet and vent out of facility.

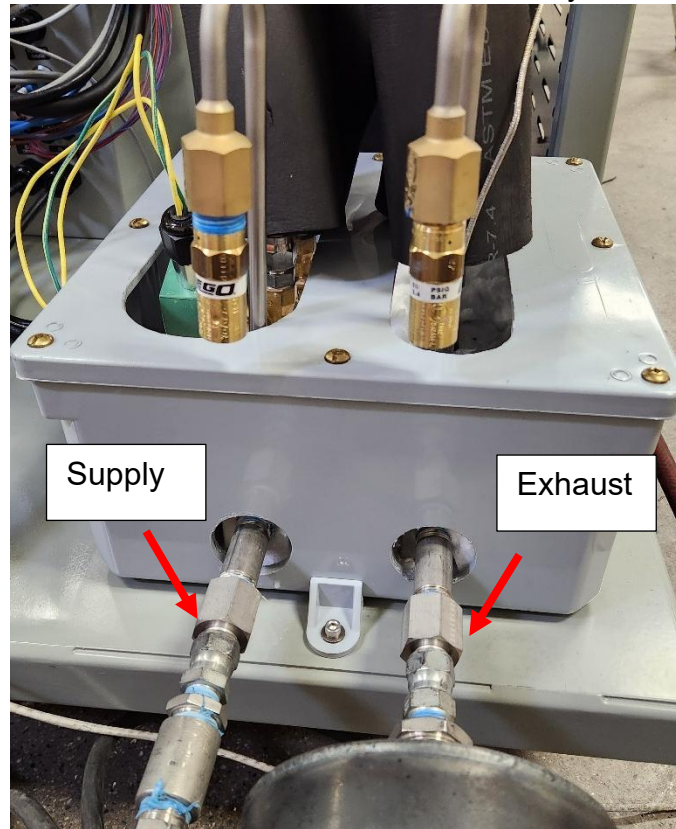


Figure 5-3. Vacuum System

DANGER

Not properly venting LN2 exhaust could cause asphyxiation and be life threatening.

5.7. Machine Start-up

NOTICE

Before starting the system, it is important to become familiar with Operation Procedures. Only trained, qualified, authorized, personnel should operate this system.

Perform the following sequence of events in the order listed for electrical re-energization of the system:

1. Ensure that all hand tools are removed from the equipment and that it is ready for start-up.
2. Notify personnel in the area that you are going to start-up the equipment.
3. Open the lock and remove the locking energy isolation device (lock) from the hole in the actuating handle of the Main Disconnect Switch.
4. Move the actuating handle to the "ON" (up) position.
5. The system will go through a 2-4-minute startup process.
6. At the end of the startup process, the touchscreen and system status lights will indicate the status of the system.

5.8. Machine Shutdown

Perform the following sequence of events in the order listed for electrical energy isolation of the system:

1. Notify personnel in the area that you are going to shut down the equipment.
2. Shutdown the system in an orderly fashion.
3. Locate the main incoming power disconnect switch-actuating handle.
4. Move the actuating handle to the "OFF" (down) position.
5. Prior to applying the energy isolation device (lock) the voltage can be verified as "OFF" by opening the door and measuring the voltage between the terminals on the bottom of the main disconnect.
6. Apply the locking energy isolation device (lock) through the hole in the actuating handle and secure the lock.
7. Verify that the tool has been isolated and de-energized by attempting to turn the main power disconnect back to the "ON" position and/or by pressing the machine start button. The machine power must not be reapplied, and/or the machine must not start.

6. Operating Procedures

6.1. Software

See HVC-3500 manual for software operation/setup instructions.

6.2. Basic Operation (High Vacuum)

High vacuum systems require two independent vacuum pumps, working together to achieve high vacuum in the chamber. The roughing pump (typically a rotary vane pump) is used to initially evacuate the chamber. The high vacuum pump takes over at a specified vacuum level (called the cross over pressure) and pumps the chamber to high vacuum. In the case of a turbomolecular high vacuum pump, the roughing pump can also act as a backing pump to evacuate the exhaust (or foreline) of the turbo pump or a dedicated backing pump can be used.

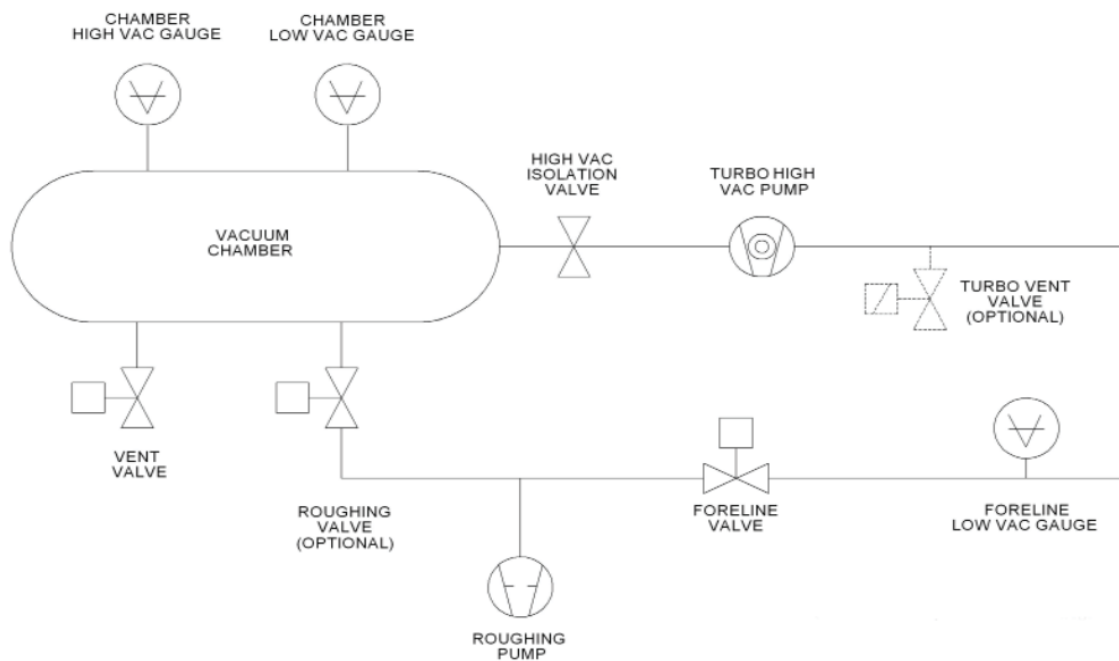


Figure 6-1. Schematic for High Vacuum System

In this configuration the turbo pump is left on and is isolated from the chamber with a valve so that the chamber may be vented to atmospheric pressure.

- 1) Start up the turbo pump and ensure it is operating at full speed with the foreline valve open and the high vacuum valve closed.
- 2) With chamber lid or door closed, close the foreline valve, then open the roughing valve to evacuate the chamber. On systems where there is a dedicated roughing pump (the pump function is not shared between

backing the turbo pump and roughing the chamber), then the foreline valve should remain open.

NOTE The turbo pump will normally run with the foreline valve closed for a few minutes while the chamber is being evacuated to the crossover setpoint. If the chamber rough evacuation will be several minutes or more, then separate, dedicated backing and roughing pumps should be used.

- 3) Once the crossover pressure is achieved (typically less than 0.1 Torr – consult turbo pump manufacturer manual for recommended crossover pressure), close the roughing valve, open the foreline valve, then open the high vacuum valve, in that order.
- 4) Allow the turbo pump to evacuate the chamber to the desired pressure.

NOTE Extreme care should be taken so that the turbo pump is not exposed to pressure shocks while at full speed. Never open the roughing or vent valves when the turbo pump is at full speed. The foreline valve should also remain open while the turbo pump is on.

- 5) Prior to venting the chamber to atmospheric pressure, close the high vacuum valve. Open the vent valve to vent the chamber to atmospheric pressure.

If using an oil-sealed mechanical pump as the roughing or backing pump, never turn off the vacuum pump unless the vacuum valve has first been closed (An alternate method is to close the vacuum valve and vent the vacuum line before turning off the vacuum pump.) This will minimize the chance of pump oil to be sucked into and contaminate the vacuum chamber and/or turbo pump.

6.3. Operation Guidelines (High Vacuum)

The most common problem associated with high vacuum systems is the inability to achieve the desired base pressure (vacuum level). Several factors can be the cause.

6.3.1. Improper Pump Sizing

In a high vacuum system, the high vacuum pump primarily is pumping gases absorbed onto the walls of the chamber and other components. If the volume of the chamber to be evacuated is the only consideration in selecting the size of the high vacuum pump, then the pump will not likely achieve the desired base pressure. One must also consider gas loads from outgassing (see below) as well as other sources (such as leaks or permeation) when sizing the pump.

6.3.2. Environmental Effects

Water vapor is the primary contaminant/gas in a vacuum system. The water vapor comes from the air and will absorb on the walls of the chamber. Pumping of the layers of water vapor on a vacuum chamber is a very slow process. High humidity, low chamber temperature, high surface area inside the chamber, and dirt and contamination all contribute to higher absorption of water vapor and poor vacuum performance. In general, if vacuum levels need to be below 10^{-6} Torr, then vacuum baking of the vacuum chamber is required. The vacuum baking may only be required initially (if the chamber is operated and protected from humidity) or may be required each vacuum cycle.

6.3.3. Contamination / Outgassing

In addition to water vapor, other chemicals and materials may contribute to the gas load of a vacuum chamber. Oils and lubricants, even from fingerprints, will limit the vacuum performance. Some contaminants with low vapor pressures (such as oils) may take weeks and months to fully pump away. Prevent liquids, chemicals, oils, or any contamination from entering the vacuum chamber.

6.3.4. Improper Use of Materials

Many materials contain chemicals, residues, or porous surfaces that will outgas under vacuum conditions. Most plastics and elastomers, for example, are not compatible with high vacuum systems. Though elastomer seals (such as Buna N and Viton) are acceptable for high vacuum systems, their use should be minimized. Minimize the use of plastics, porous materials, and materials with rough surfaces.

6.3.5. Improper Tooling Design

A common problem in high vacuum systems is virtual leaks. Virtual leaks are introduced into a vacuum system by improper vacuum chamber or fixture design. A virtual leak is a small pocket or void that communicates with the vacuum system through a very small path. A common example of a virtual leak is a threaded screw installed into a blind, tapped hole. The dead space below the screw in the tapped hole is the pocket and the threads form the small path. The pocket of air must be evacuated through the threads each time the chamber is vented. This can significantly slow down the evacuation of the chamber. Venting the screw or eliminating the blind hole will eliminate the virtual leak.

6.3.6. Leaks

Most often leaks are created in a vacuum system when a seal is broken (whether a door seal or a flange seal). If a leak is assumed, always check the item in the system that was last modified. Use a helium leak detector as a tool to locate leaks.

Table 6-1. DOs and DON'Ts for operating a high vacuum system

DO	DON'T
Always touch the chamber or any components that will go inside the chamber with clean, gloved hands.	Touch inside the chamber with bare hands.
Bake out the chamber to achieve a deeper vacuum.	Place plastics or porous materials in a vacuum chamber.
Clean any part prior to placing into a vacuum chamber.	Use oils or greases inside the vacuum chamber unless they are high vacuum compatible. Then use sparingly.
Vent the vacuum chamber with dry nitrogen to prevent water vapor from absorbing on the clean walls of the chamber.	Allow parts with virtual leaks into a vacuum chamber.
Keep the chamber under vacuum at all times, even when not in use.	Use dirty tools inside the vacuum chamber

7. Faults and Troubleshooting

See HVC-3500 manual for information regarding the HVC-3500 controller and software.

7.1. Vacuum Chamber Troubleshooting

Problem	Cause	Solution
Chamber will not pump down	Leak in vacuum line	Check hose, hose connections, and clamps
	Door and gasket not sealing flat	Adjust hinge
	Defect in gasket	Replace gasket. See spare parts section.
	Defect in pressure relief valve	Replace pressure relief valve
Chamber evacuates slowly or will not achieve base pressure	Poor vacuum pump performance	Change pump oil Vacuum pump needs rebuild
	Outgassing of product	Minimize outgassing effect, remove offending part
	Outgassing of vacuum chamber	Clean vacuum chamber and components
	Leak in system	Inspect all components and connections. Helium leak test chamber system, if possible. Perform rate of rise test to see if you have a leak or outgassing problems. *
	Defective vacuum gauge	Check operation of gauge
Heaters won't turn on	Loose wiring	Check quality of electrical connections
	Faulty Heater	Replace heater. See spare parts section.
LED lighting won't turn on	Loose wiring	Check quality of electrical connections
Controller not reading temperature	Loose thermocouple wire	Check thermocouple wire connections

* A rate of rise test can be performed by evacuating the chamber to the lowest pressure practical, then closing all valves and plotting the pressure rise

over time up to pressures over 1 Torr. If the pressure rise is linear then the culprit is likely an external leak. If the pressure rise slows down at higher pressures (is non-linear), then the culprit is likely outgassing.

7.2. Software Faults

Whenever a fault occurs refer to the table below for help in identifying faults as well as their respective likely causes. There are two kinds of faults 1) global or system faults and 2) process related faults. Global faults can occur at any time whereas process faults only occur at certain steps in the test process that do not function properly. Note that fault histories can be reviewed on the fault history page.

When a fault occurs, the control system stops all processes. To recover from a fault, press the red Abort/Reset button. The system will then proceed through the recovery process back to a ready-to-test state.

Check fault list and remove/add as needed

Fault Description	Fault Reason	Fault Troubleshooting
Fault: Control Air Low	Air pressure below setpoint on main air pressure switch at any time.	• Air source turned off or not connected. • Leak in air lines. • Main air regulator needs adjustment.
Fault: Cycle Aborted	The test was terminated, at user request, by pressing the Abort/Reset button.	• Press Abort/Reset, again, to reset the fault and recover the system.

8. Recommended System Maintenance and Spare Parts

This section summarizes the recommended maintenance procedures and spare parts based on LACO's experience and understanding of the Vacuum Control System environment and is not meant to replace recommendations of original equipment manufacturers. Refer to their equipment documentation for more detailed maintenance and operation instructions.

LACO Technologies offers a variety of service agreements so that companies can have assistance in maintaining their vacuum systems. Contact LACO Technologies with any maintenance or service agreement questions.

Refer to the Spare Parts list in this section for a comprehensive list of all spare parts. Take note of all parts noted as critical spares and all recommended replacement intervals.

8.1. General Cleaning

The system should be kept clean, dry, and free of dust, dirt, debris. As needed, use a standard cleaner such as simple-green and a dry cloth to clean outside surfaces.

8.2. Vacuum Chamber

Regularly clean the vacuum chamber to keep the vacuum process running at optimum performance. A dirty chamber may adversely affect the vacuum process, the vacuum pump, chamber gasket, or other vacuum components. Clean the chamber with mild detergent and rinse with DI water. Methanol or other mild solvents can also be used to clean or wipe down the chamber. Use lint free wipes to wipe out the chamber.

8.3. Vacuum Pump

The provided pump does not use oil in its vacuum circuit, but does require oil for operation. Oil should be changed regularly to maximize pump performance and life. Refer to the pump manufacturer's manual for specifics on changing the pump oil and other pump maintenance.



WARNING

If the pump has been used on corrosive, toxic, or volatile chemicals, observe proper safety precautions before removing the drain plug.

8.4. Seals

Inspect the chamber gasket regularly to ensure no defects or wear have occurred. If cracks begin to appear in the gasket, replace it immediately. See the spare parts section for ordering replacement gaskets. Do not clean the gasket with any solvents as this will shorten the gasket life. To clean the gasket, wipe it down with a lint free wipe and DI water. A suitable vacuum grease (e.g. Krytox LVP) can be applied to the gasket to help enhance gasket. Apply the grease sparingly to the gasket to achieve a smooth, thin film.

Seal materials degrade over time, so replacing all seals annually is recommended.

8.5. Component Maintenance

See individual component documentation for component maintenance.

8.6. Gauge Calibration

LACO recommends calibration of all vacuum gauges annually or as required by the user's quality policy

LACO provides NIST Traceable calibration services for all types of leak standards and vacuum gauges. Our A2LA Accredited calibration lab complies with ISO/IEC 17025-2005, ANSI/NC SL Z540-1-1994, and with the ISO-9001 quality standard. LACO's responsive technical support staff ensures fast delivery of a broad calibration range that encompasses all leak standard brands and vacuum gauge types.

8.7. Spare Parts

NOTICE

Take note of all parts noted as critical spares and all recommended replacement intervals in this section.

Items identified as critical spares are for preventative maintenance, have a high likelihood of needing replacement, or may have a long lead time. Parts listed that are not critical spares may fail, but their failure is less likely than critical parts. Take special note of the replacement intervals in the tables below as it is always best to replace a component before it fails and leads to equipment downtime.

The following is a list of spare parts that can be kept on hand by the user for regular maintenance and repair. This list is not all-inclusive. Contact LACO with any questions regarding spare parts for this system.

LACO PART #	DESCRIPTION	REPLACEMENT INTERVAL	CRITICAL SPARE	QTY/ SYSTEM
LGO30V	CHAMBER GASKET	ANNUALY	X	1
Y5828	DESICCANT BREATHER	AS NEEDED INDICATED BY COLOR CHANGE		1
LVOF256	PFPE PUMP OIL	PER SUPPLIED PUMP MANUAL DEPENDING ON USAGE	X	1
IN3MA0006000N-C	HIGH VACUUM GAUGE	AS NEEDED, CALIBRATE ANNUALLY	X	1
IN3PC10010001-C	ROUGH VACUUM GAUGE	AS NEEDED, CALIBRATE ANNUALLY	X	1
IN3PI10012001	FORELINE SETPOINT GAUGE	AS NEEDED		1
LVVRAN25PNS	NW25 PNEUMATIC VALVES	AS NEEDED		3
LVVRAN40PNS	NW40 PNEUMATIC VALVES	AS NEEDED		1
Y4449	0.2 PSI SYSTEM RELIEF VALVE	AS NEEDED		1
SVSS8VCR2	½" VCR GASKET	REPLACE EVERY DISCONNECTION	X	10
YA4557	3/8" LN2 CRYO VALVE	AS NEEDED		2
YA5306	50 PSI LN2 RELIEF VALVE	AS NEEDED		2
NAHD40-2	VACUUM ROUGHING PUMP	AS NEEDED		1
LE830071V2000	HIGH VACUUM TURBO PUMP	AS NEEDED		1
LMSA119198	SYSTEM INTERNAL LEDS	AS NEEDED		1
Y137831	THERMAL PLATEN	AS NEEDED		1
Y138114-Y138117	FLEXIBLE SHROUD HEATERS	AS NEEDED		1 EA

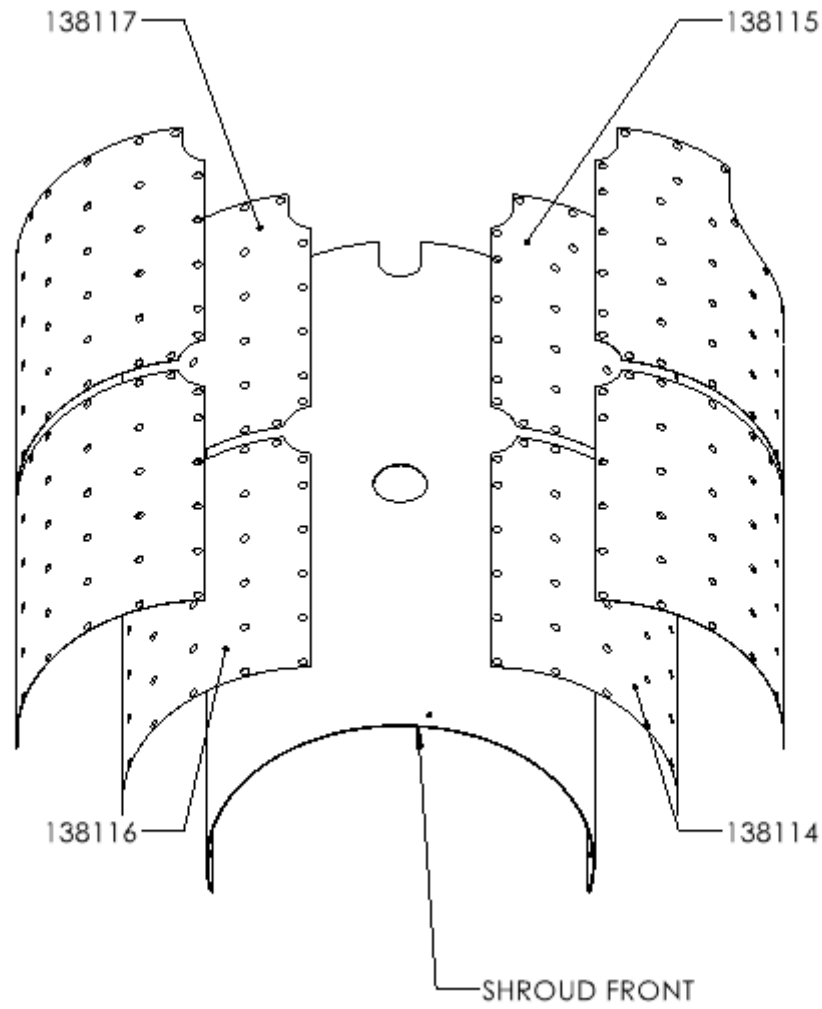


Figure 8-1. Heater Locations